

Raw Sketch: Safety Communication Protocols

Industrial Communication Standards | Advanced Embedded Systems

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Technical Scope

This paper compares the five largest functional-safety communication protocols deployed over standard Industrial Ethernet networks. All five are governed by IEC 61784-3 [1] and implement the Black Channel principle, making them directly comparable on the same architectural and performance axes.

#	Protocol	Transport	Lead Organisation	Primary Market
1	PROFIsafe	PROFINET IRT/RT	Siemens / PI	Global, process and discrete
2	FSoE	EtherCAT	Beckhoff / ETG	Europe, motion control
3	CIP Safety	EtherNet/IP	ODVA / Rockwell	North America, discrete
4	openSAFETY	POWERLINK / Modbus	B&R / EPSG	Europe, machine building
5	CC-Link IE Safety	CC-Link IE Field	Mitsubishi/CLPA	Asia-Pacific, automotive

Out of scope: ISO 26262 in-vehicle safety, wireless safety protocols, legacy fieldbus safety (PROFIBUS, DeviceNet Safety), and IO-Link Safety.

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1. Abstract

- Problem statement: industrial machinery poses severe risk of human injury when safety-critical signals (E-Stop, light curtains, robot guards) are lost, delayed, or corrupted.
- Introduce the Black Channel concept as the shared architectural response across all five protocols [1].
- Paper contribution: compare five IEC 61784-3 protocols [1] on architecture, performance, certification pathway, and application suitability.
- Key finding preview: all five reach SIL 3 / PL e [2][3][4]; differentiators are determinism, ecosystem breadth, regional deployment, and certification overhead.
- Scope note: factory automation only; 6-page limit; IEEE two-column format.

2. Introduction

2.1 Why Functional Safety Matters in Industrial Systems

- Definition: functional safety is the part of overall safety that depends on the correct operation of electrical, electronic, and programmable systems [2].
- Real consequences of failure: robot arms striking operators, press brakes closing on hands, conveyors running during maintenance; all caused by missed or delayed safe-state commands.
- Scale: machinery accidents account for approximately 20 % of fatal workplace injuries in the EU (Eurostat); North American OSHA data reflect similar proportions.
- SIL framework: risk quantified as Probability of dangerous Failure per Hour (PFH); SIL 3 requires $PFH < 10^{-7}/h$, the target for most machine-guarding and robot-safety applications [2].
- Regulatory driver: IEC 62061 [3] and ISO 13849 [4] mandate certified safe communication for Category 3 / PL d and above machine designs.

2.2 Functional Safety Requirements Specific to Communication

- Standard Ethernet was not designed with safety in mind: no guaranteed delivery, no bounded latency, no inherent protection against transmission errors.
- IEC 61784-3 [1] defines seven error classes every certified protocol must address:
 - Corruption; Unintended repetition; Incorrect sequence; Loss; Unacceptable delay; Insertion; Masquerading.
- Process Safety Time (PST): the application-level deadline within which a safe state must be reached after hazard detection, typically 10–200 ms for machine-guarding applications [2].
- Black Channel principle: the safety PDU self-protects against all seven error classes without requiring a certified transport layer, enabling deployment over any standard Industrial Ethernet [1].

2.3 Motivation for a Five-Protocol Comparison

- No single protocol dominates globally; the market is regionally fragmented by vendor ecosystem [9][12].
- PROFIsafe: >35 million safe nodes (2023) [13]. FSoE: benchmark for motion-control determinism [5]. CIP Safety: North American automotive standard [6]. openSAFETY: the only royalty-free open specification [7]. CC-Link IE Safety: Japanese automotive supply-chain standard [8].
- A practitioner selecting a safety architecture must understand trade-offs across all five; this paper provides that comparative foundation.

2.4 Paper Structure

- Section 3: background and evolution of safety communication standards.
- Section 4: protocol architectures and comparison framework.
- Section 5: evaluation across performance, certification, and application KPIs.
- Section 6: conclusions and protocol selection guidance.

3. Literature Review / Background

3.1 Evolution: Fieldbus Safety to Industrial Ethernet Safety

- 1980s–1990s: PROFIBUS DP, DeviceNet, CC-Link: deterministic, proprietary physical layer; safety via dedicated wiring or proprietary sublayers [9][12].
- 2000s: shift to Industrial Ethernet (PROFINET, EtherCAT, EtherNet/IP, POWERLINK, CC-Link IE): higher bandwidth, IT-compatible, but not deterministic by default [11][12].
- 2005–2012: safety sublayers retrofitted on top of each variant; all independently converge on the Black Channel architecture [1].
- IEC 61784-3:2010 (revised 2021): first standard to unify requirements for all safety protocol families under a single Communication Profile Family (CPF) model [1].

3.2 The Black Channel Principle

- The standard transport is treated as untrusted: no safety assumption is made about delivery [1].
- The safety PDU is self-contained: CRC, consecutive number, node/connection ID, and watchdog timer; together sufficient to detect all seven IEC 61784-3 error classes [1].
- Transport-agnostic design: the same safety application layer can run over PROFINET, Modbus TCP, or serial, enabling future migration to TSN transports without modifying safety logic [1].

3.3 Governing Standards

- IEC 61508 [2]: umbrella functional safety standard; defines SIL 1–4 and PFH targets; sets hardware and software development lifecycle requirements.
- IEC 62061 [3]: machine-specific safety; defines SIL requirements for Safety-Related Control Functions (SRCF).
- ISO 13849-1 [4]: Performance Level (PL a–e) framework; used in parallel with IEC 62061 by machine builders.
- IEC 61784-3 [1]: assigns each protocol to a CPF: CP 3/1 (PROFIsafe), CP 12/1 (FSoE), CP 2/1 (CIP Safety), CP 13/1 (openSAFETY), CP 8/1 (CC-Link IE Safety).
- IEC 61511: process-industry safety instrumented systems; relevant for PROFIsafe deployments in oil/gas and chemical plants [2].

4. Protocol Architectures: Comparison Framework

4.1 Per-Protocol Architecture Overview

PROFIsafe (CP 3/1) [1][13]

- F-Host / F-Device model; F-PDU inserted transparently into PROFINET IO data slots.
- CRC-24 polynomial; 8-bit F-Address combined with a consecutive number for sequencing.
- Watchdog configured via F_WD_Time parameter; F-Supervisor in F-Host drives passivation.
- Engineering tool: TIA Portal Safety (Siemens); device catalogue exceeds 2 000 certified products.

FSoE: Fail Safe over EtherCAT (CP 12/1) [1][5]

- Master/Slave model; FSoE frame structure: Command Byte | User Data | Connection ID | CRC.
- CRC-8 for payloads ≤ 2 bytes; CRC-16 for larger payloads; 2-byte Safety Header Block (SHB) for sequencing.
- Slave self-passivates on watchdog expiry; no master command required.
- Most compact safety overhead (~ 6 bytes); lowest achievable cycle time among the five.
- Engineering tool: TwinCAT Safety (Beckhoff).

CIP Safety (CP 2/1) [1][6]

- Originator/Target roles mapped over the standard EtherNet/IP connection model.
- Dual-packet format: actual data transmitted alongside its bit-complement copy for cross-checking.
- Time Coordination Object handles network delay compensation explicitly.
- Safety Supervisor Object (class 0x39) manages the safety state machine.
- Highest protocol overhead (~ 12 bytes) but best interoperability with existing EtherNet/IP infrastructure.
- Engineering tool: Studio 5000 / GuardLogix (Rockwell Automation) [14].

openSAFETY (CP 13/1) [1][7]

- Frame-in-frame design: safety frame embedded within a standard POWERLINK or Modbus TCP payload; the transport layer treats it as ordinary data.
- SPDO (Safety Process Data Object): two independent sub-frames, each carrying its own CRC (CRC-8 + CRC-16); both must fail to trigger passivation.
- The only royalty-free, fully open specification among the five.
- Engineering tool: Automation Studio Safety (B&R) [15].

CC-Link IE Safety (CP 8/1) [1][8]

- Safety Communication Layer (SCL) mapped into CC-Link IE Field cyclic frames.
- CCITT CRC-16; node address combined with a frame counter for sequencing.
- Passivation coordinated centrally by a Mitsubishi MELSEC iQ-R safety PLC.
- Constrained to ring/line topology; ecosystem tightly coupled to Mitsubishi hardware.
- Strongest presence in Japanese automotive manufacturing; aligned with JIS B 9961.
- Engineering tool: GX Works3 (Mitsubishi Electric).

4.2 Architecture Comparison

Feature	PROFIsafe	FSoE	CIP Safety	openSAFETY	CC-Link IE
IEC 61784-3 CPF [1]	CP 3/1	CP 12/1	CP 2/1	CP 13/1	CP 8/1
Max payload (bytes)	123	126	Variable	8 (SPDO)	8
CRC scheme	CRC-24	CRC-8/16	CRC-S1 (16b)	CRC-8+CRC-16	CRC-16
Sequence mechanism	F-Addr+consec.	2-byte SHB	Conn. serial	Dual sub-frame	Frame counter
Safety overhead (B)	~ 4	~ 6	~ 12	~ 11	~ 6

Feature	PROFIsafe	FSoE	CIP Safety	openSAFETY	CC-Link IE
Topology	High	Medium	High	Medium	Low (ring)
Open specification?	No	No	No	Yes [7]	No

4.3 Fail-Safe Passivation Mechanism

- All five: configurable watchdog timeout; expiry causes outputs to be set to the safe state [1].
- PROFIsafe: F-Supervisor in F-Host initiates passivation of F-Device outputs [1][13].
- FSoE: slave self-passivates independently; master issues a Reset command to re-enable [5].
- CIP Safety: target enters safe state autonomously; no command from originator required [6].
- openSAFETY: passivation requires failure of both sub-frame CRCs, reducing nuisance trips [7].
- CC-Link IE Safety: SCL watchdog triggers; central safety PLC coordinates passivation across all nodes [8].

5. Evaluation

5.1 Performance KPIs

- **Latency:** one-way time from safe input event to safe output reaction (ms).
- **Jitter:** cycle-to-cycle variance in safety data delivery (μ s).
- **Minimum cycle time:** fastest guaranteed refresh interval of safety data.
- **Safety reaction time:** worst-case time to reach safe state, typically 2–3× the cycle time.
- **PFH:** Probability of dangerous Failure per Hour; SIL 3 requires $PFH < 10^{-7}/h$ [2].
- **Throughput (not tabulated):** throughput for safety protocols is the product of payload size and cycle frequency; since both are already captured in the cycle time and safety overhead rows, raw throughput adds no discriminating power to this comparison. All five protocols operate over 100 Mbps or 1 Gbps Industrial Ethernet, meaning the channel is never the bottleneck. The binding constraint is always timing, not bandwidth.

5.2 Performance Comparison

Metric	PROFIsafe	FSoE	CIP Safety	openSAFETY	CC-Link IE
Min. cycle time	~1 ms (IRT)	~125 μ s	~1-10 ms	~400 μ s	~500 μ s
Typical jitter	<1 μ s (IRT)	<1 μ s	~100 μ s	<1 μ s	~1 μ s
Safety reaction time	~4-8 ms	~1-2 ms	~8-20 ms	~2-4 ms	~4-6 ms
Max certified SIL	SIL 3	SIL 3	SIL 3	SIL 3	SIL 3
PFH ceiling	< $10^{-8}/h$	< $10^{-8}/h$	< $10^{-8}/h$	< $10^{-8}/h$	< $10^{-8}/h$

Sources: ETG.5100 [5], Siemens SIMATIC Safety manual [13], Rockwell 1756-RM012 [14], B&R openSAFETY white paper [15], CLPA CC-Link IE Safety specification [8].

5.3 Certification and Reliability Considerations

- All five require assessment by an accredited third-party body (TÜV SÜD, TÜV Rheinland, or equivalent) in accordance with IEC 61508 Part 3 [2].
- Hardware certification: FMEA, diagnostic coverage (DC), common-cause failure (CCF) analysis; hardware fault tolerance (HFT) must match the target SIL [2].
- Software certification: IEC 61508 Part 3 software lifecycle; MISRA C compliance is standard practice for embedded safety stacks [2].
- Ongoing reliability: device MTBF and proof-test interval requirements must be met, with a typical interval of 1–20 years for SIL 3 devices [2].
- openSAFETY advantage: as a royalty-free open specification, any vendor may implement and independently certify the stack [7]; all other protocols require licensing from the owning consortium.
- CC-Link IE Safety: TÜV certificates are issued under JIS B 9961, which is internationally recognised but introduces additional administrative steps for customers outside Japan [8].

5.4 Application Area Suitability

- FSoE / EtherCAT [5]: high-axis-count motion control, servo press, CNC, collaborative robot cells; lowest latency is the decisive factor.
- openSAFETY / POWERLINK [7]: synchronised multi-axis European OEM machinery; machine builders seeking a vendor-independent, royalty-free safety stack.
- PROFIsafe / PROFINET [11][13]: large mixed installations combining sensors, drives, and PLCs on a single network; process industry (oil/gas, chemical); widest global device catalogue.
- CIP Safety / EtherNet/IP [6][14]: North American automotive and discrete manufacturing; brownfield EtherNet/IP sites; strongest interoperability with non-safety EtherNet/IP devices on the same network.
- CC-Link IE Safety [8]: Japanese and broader APAC automotive Tier 1/Tier 2 suppliers; Mitsubishi MELSEC environments.

6. Conclusions

6.1 Summary of Findings

- All five protocols achieve SIL 3 / PL e under IEC 61508 [2] and IEC 62061 [3]; the safety ceiling is not a differentiating factor.
- Performance: FSoE [5] and openSAFETY [7] achieve sub-millisecond cycle times and sub-microsecond jitter; CIP Safety [6] lags on determinism but leads on network integration flexibility.
- Ecosystem: PROFI-safe [13] holds the largest global installed base and broadest certified device catalogue.
- Openness: openSAFETY [7] is the only royalty-free specification, lowering the barrier for new device vendors.
- Regional dominance: CC-Link IE Safety [8] is the standard in Japanese automotive manufacturing; CIP Safety [6] holds equivalent standing in North American automotive.

6.2 Protocol Selection Guidance

- **FSoE [5]:** select when motion control performance and sub-2 ms reaction time are required and EtherCAT is already deployed.
- **PROFI-safe [13]:** select for large mixed installations, process-industry applications, or where Siemens infrastructure and the widest device choice are priorities.
- **CIP Safety [6]:** select for North American plants with Rockwell / Allen-Bradley PLCs and existing EtherNet/IP brownfield networks.
- **openSAFETY [7]:** select for European OEM machine builders requiring vendor independence and an open certification path.
- **CC-Link IE Safety [8]:** select for APAC automotive environments built on Mitsubishi MELSEC infrastructure.

6.3 Future Work

- Impact of IEEE 802.1 TSN on determinism guarantees across all five; all Black Channel protocols can migrate to TSN transports without modifying the safety sublayer [1].
- Protocol convergence under IEC/IEEE 60802 (TSN profile for Industrial Automation).
- IO-Link Safety as a complementary sensor-level layer operating beneath all five protocols [1].

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